UNIVERSITY OF OXFORD

Finding Resonance:

A Tool for Singers and Phoneticians to Explore Formant Tuning

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BAAP | UNIVERSITY OF WARWICK



Image: Andesana with Lambdoma | Anna-Maria Hefele

Resonance Basics

(1/2)

Glottis = source

Fourier transform → fundamental + harmonics (multiples of f0) + noise

Fixed cavity resonators / **filters**

standing wave (Pavarotti/Mariah Carey note in the shower)

Fixed mechanical resonators in the body

sympathetic vibration → felt buzzing (cheek bones, rib cage)

Vocal tract = **malleable** cavity resonator / **filter**...

we can choose the frequencies to amplify

Resonance Basics

(2/2)

Source-filter theory

Resonance near harmonic:

amplification, aesthetically-pleasing, vocal ease!

BUT...

Non-linear coupling effects... the filter *can* affect the source

Resonance near harmonic:

f0 discontinuities, instability!

Titze 2008; Maxfield et al. 2017

Talk Overview

- American accents in pop music (default ⟷ effort)
- Continuum from Singy to Speaky
- Formant-tuning (formant-harmonic alignment)
- ‘Finding Resonance’ tool intro

American accents in pop music (1/8)

Geographic place – primary structuring variable for spoken dialects

Musical genre – primary structuring variable for singing accents

Pop Song English (PSE) (Gibson 2024) supralocal norm in a wide range of genres. But lots of other singing accents:

- hip hop (Alim et al. 2009)
- punk (Trudgill 1983, Coupland 2011)
- Britpop (Simpson 1999)
- country (Duncan 2017)
- reggae (Westphal 2018, Gerfer 2018)
- folk (Watts & Andres Morrissey 2019)
- choral music (Wilson 2017)

What happens automatically? ⟷ What requires more intention?

American accents in pop music

Cross-Situational Learning!
Xinbing Luo et al (BAAP 2026)

Scenario 1

Default = Own accent
Effort = ‘American’

Effortful (& selective) adoption of American features in song

Trudgill 1983, Simpson 1999

Scenario 2

Default = Pop Song English
Effort = Own accent

Effortful (& selective) adoption of own accent features in song

Beal 2009; Gibson & Bell 2012; Gibson 2024; Gibson & Hay 2025

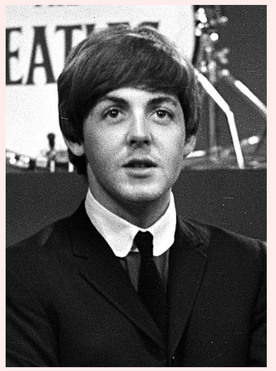
American accents in pop music (3/8)

Scenario 1

Default = ‘own accent’
Effort = ‘American’

Trudgill 1983, Simpson 1999

How to diagnose automatic vs. intended?
Overshoot = evidence of intention (Bell & Gibson 2011)



There were birds
In the sky
But I never saw/r/ them winging

American accents in pop music (4/8)



Dylan Storey
NZ singer-songwriter

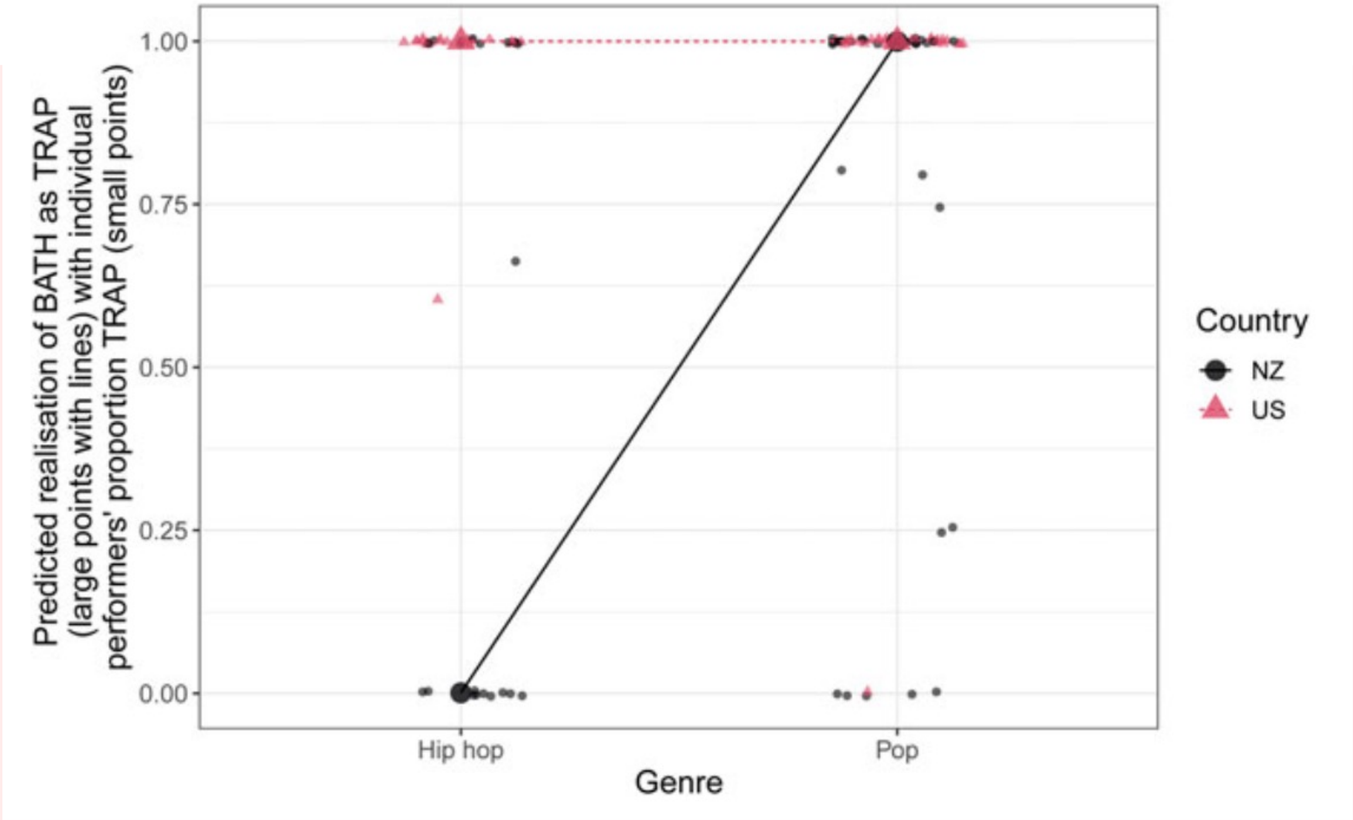
- Singing American accent: **easy**
- Singing NZ accent: “**difficult** ... *you have to be really conscious*”

Scenario 2

Default = Pop Song English
Effort = Own accent

Beal 2009; Gibson & Bell 2012;
Gibson 2024; Gibson & Hay 2025

American accents in pop music



Categorical/‘decision-point’ variable (TRAP-BATH split)
Somer artists avoid ‘American’ variant

Gibson 2024

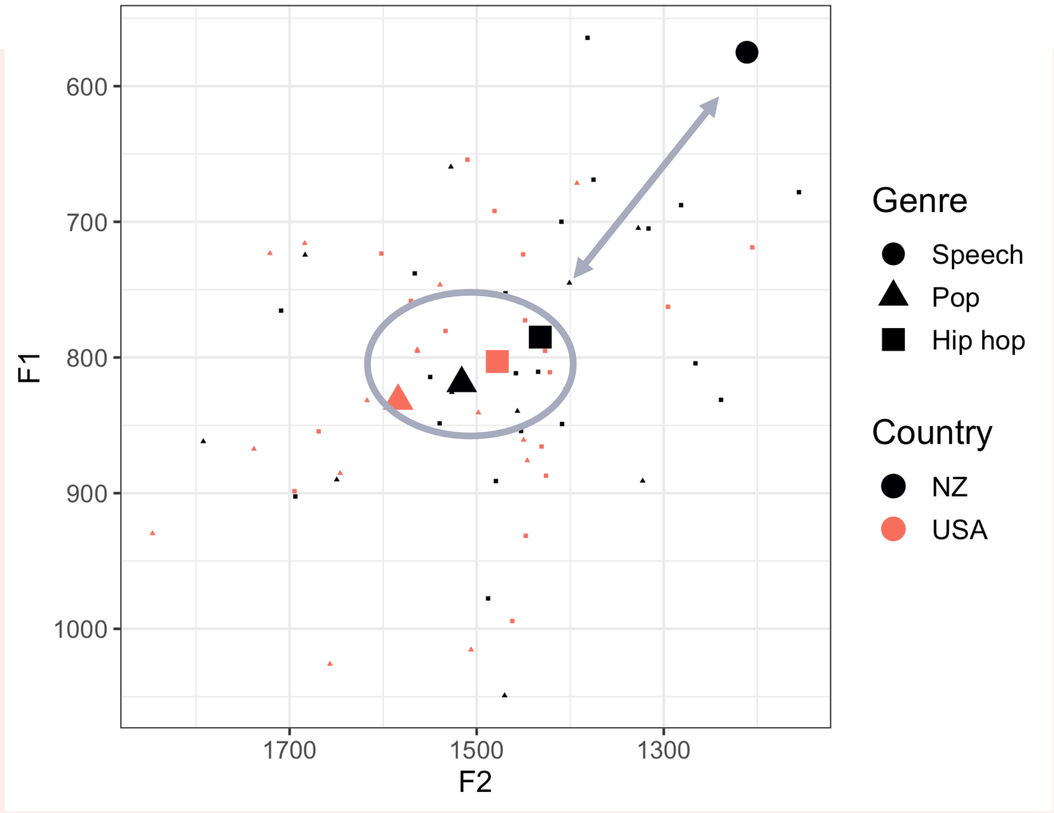
Scenario 2

Default = Pop Song English

Effort = Own accent

Beal 2009; Gibson & Bell 2012;
Gibson 2024; Gibson & Hay 2025

American accents in pop music



Gradient variable (LOT)

Pop Song English for most singers/rappers
including hip hop artists who want to project local identity

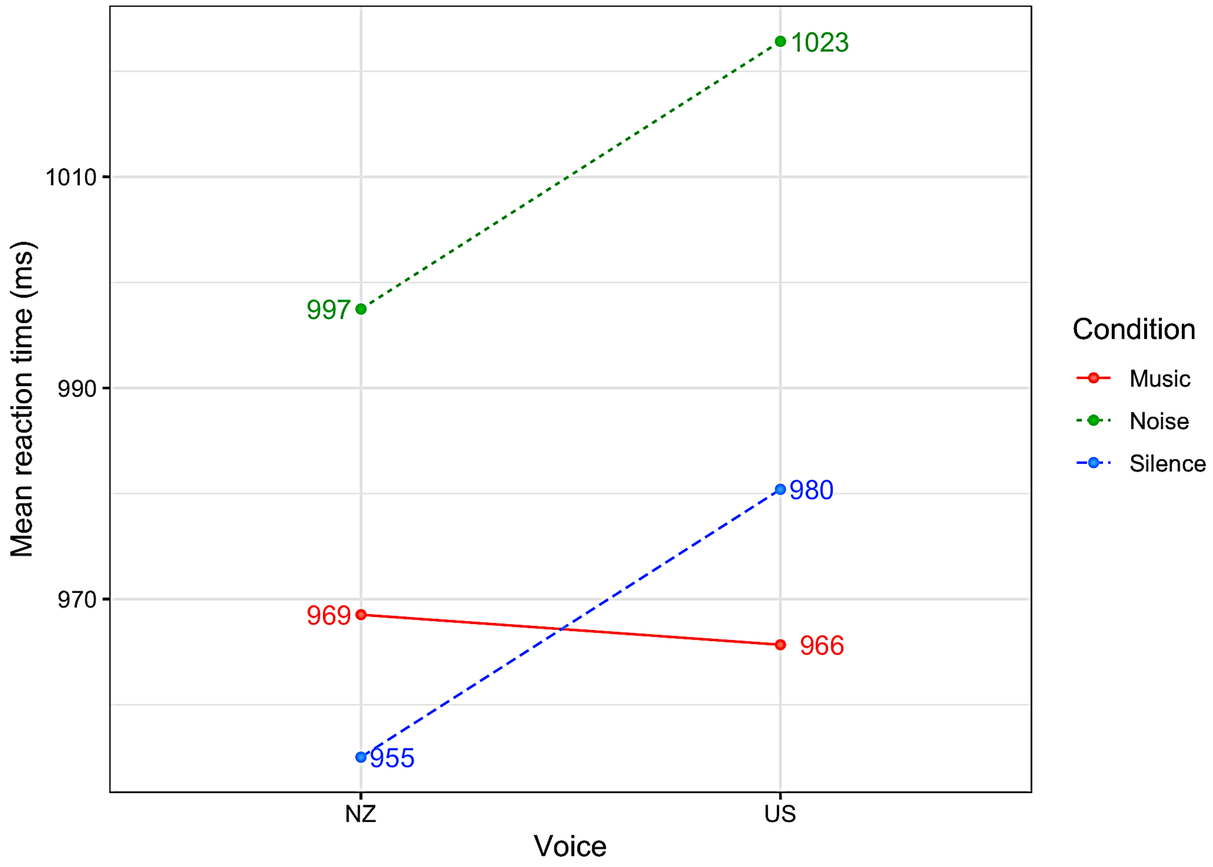
Scenario 2

Default = Pop Song English

Effort = Own accent

Beal 2009; Gibson & Bell 2012;
Gibson 2024; Gibson & Hay 2025

American accents in pop music



NZ listeners, lexical decision:
Faster RT to US voice in music

Gibson & Hay 2025

Scenario 2

Default = Pop Song English

Effort = Own accent

Beal 2009; Gibson & Bell 2012;
Gibson 2024; Gibson & Hay 2025

American accents in pop music (8/8)

Scenario 1

Default = Own accent
Effort = ‘American’

Effortful (& selective) adoption of American features in song

Trudgill 1983, Simpson 1999

Beyond accent/dialect: Singing is also ... SUNG!

We need to consider singing technique effects

Role of resonance in singers’ vowel placement

Andres Morrissey 2008; Gibson 2010

Scenario 2

Default = Pop Song English
Effort = Own accent

Effortful (& selective) adoption of own accent features in song

Beal 2009; Gibson & Bell 2012; Gibson 2024; Gibson & Hay 2025

Continuum from Singy to Speaky (1/3)

Speech ⟷ Speaky singing (e.g. rap) ⟷ Singy singing (e.g. opera)

shorter vowels	⟷	longer vowels
low f0		high f0
mobile f0		stable f0
lower F1		higher F1
		(vowel opening and/or larynx raising)
less resonant?		more resonant?

Continuum from Singy to Speaky (2/3)

Within singing: higher f0 associated with higher F1

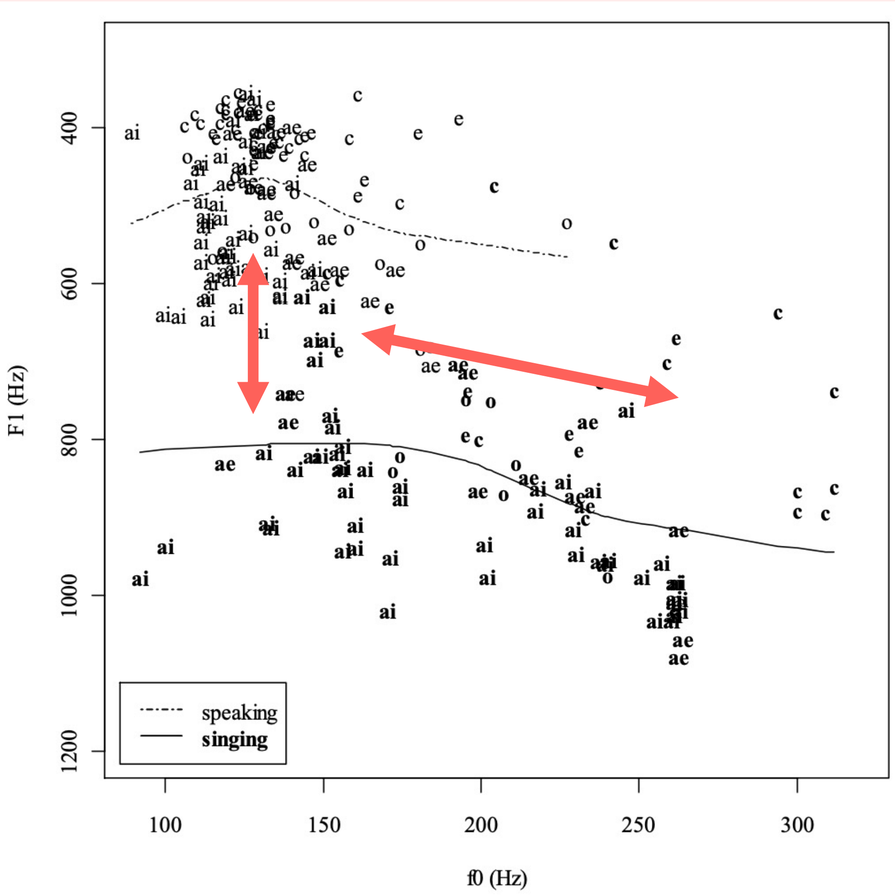
“the need to open the mouth more for high notes is somehow “natural” and does not have to be learned or taught”

(Austin, 2007, p. 78).

F1 higher in singing than speech (dialect effect)
F1 correlated with f0 in singing (singing technique)

(Gibson 2010)

Dialect effect > singing technique effect





Continuum from Singy to Speaky (3/3)

Singiness within a song:

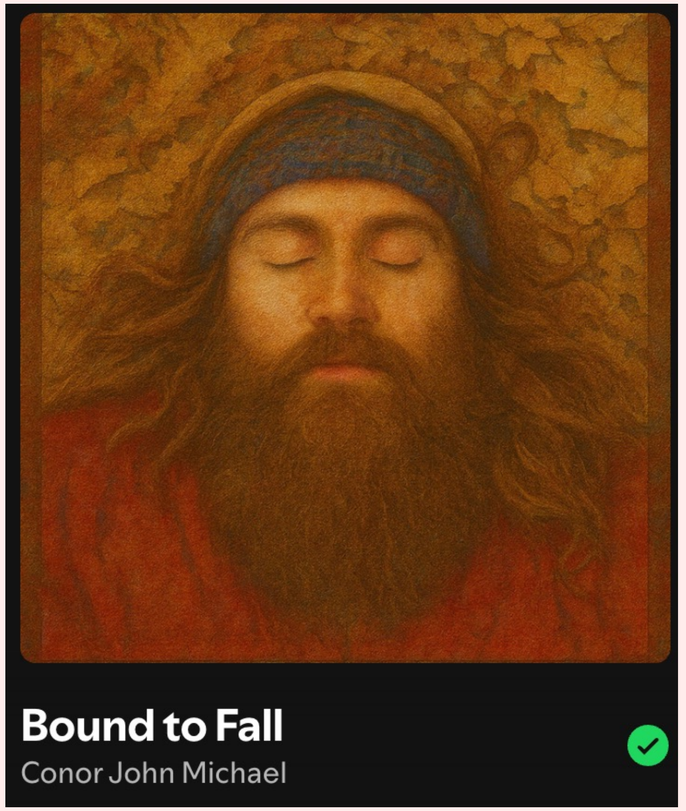
Speaky verses: (GOOSE-fronting)

“too [ʉː] many intrusive thoughts”

Singy choruses: (GOOSE-backing for resonance reasons?)

“bound to [uː] fall”

More singy = further from speech (Simpson 1999, p.360)



Conor John Michael (South London)

Formant-tuning (formant-harmonic alignment)

Overtone singing:

the singiest singing of all!

Harmonics can be heard as notes

Extreme formant- tuning!
Resonances align with harmonics

Here: bunching F2/F3 Titze & Maxfield 2015

What about ‘normal’ singing?



Anna-Maria Hefele (@annamariahefele)

Formant-tuning (formant-harmonic alignment) (2/2)

Classical sopranos: keep F1 >= f0... *‘open mouth on high notes’* (Joliveau et al. 2004)

Musical theatre: align F1 to H2 to achieve the belt timbre (Herbst et al., 2023)

Bulgarian folk singers (Henrich et al. 2007); Connected speech (Story et al., 2025)

- How to account for singing technique effects in sociophonetics of singing?
- First step... a tool to experiment and explore... Finding Resonance

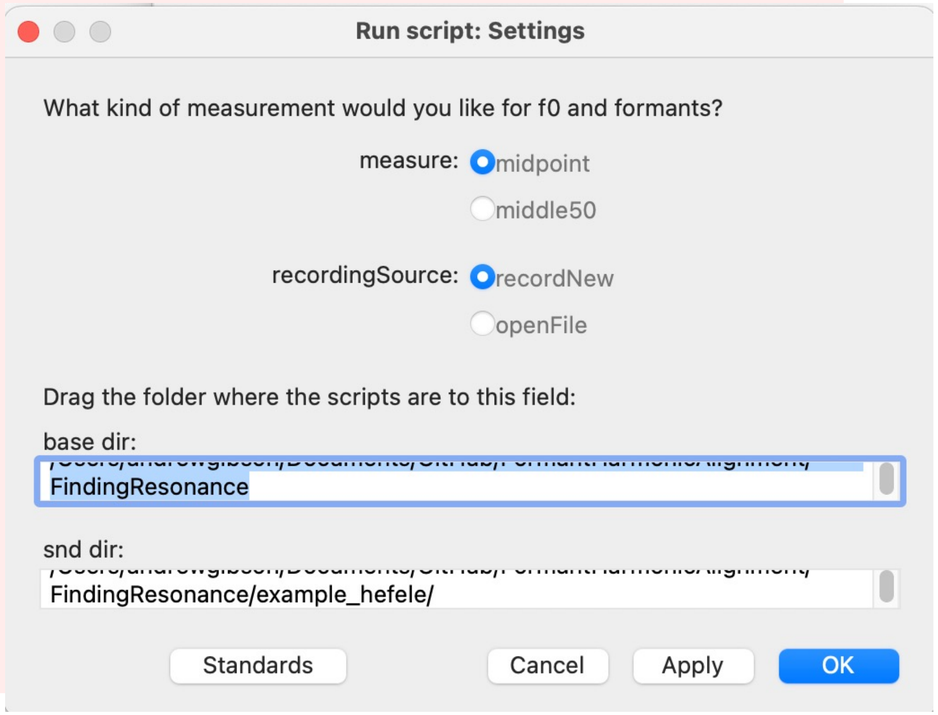
Finding Resonance (beta):

github.com/andygibsonling/findingresonance

Finding Resonance: Praat script

Get soundfiles for target vowels

- Interface to record new sounds with Praat, or to upload soundfiles
- Optionally take multiple clips per soundfile
- Saves long sound, short sounds, spectra and csv



Finding Resonance: Praat script

Get measures for each soundfile

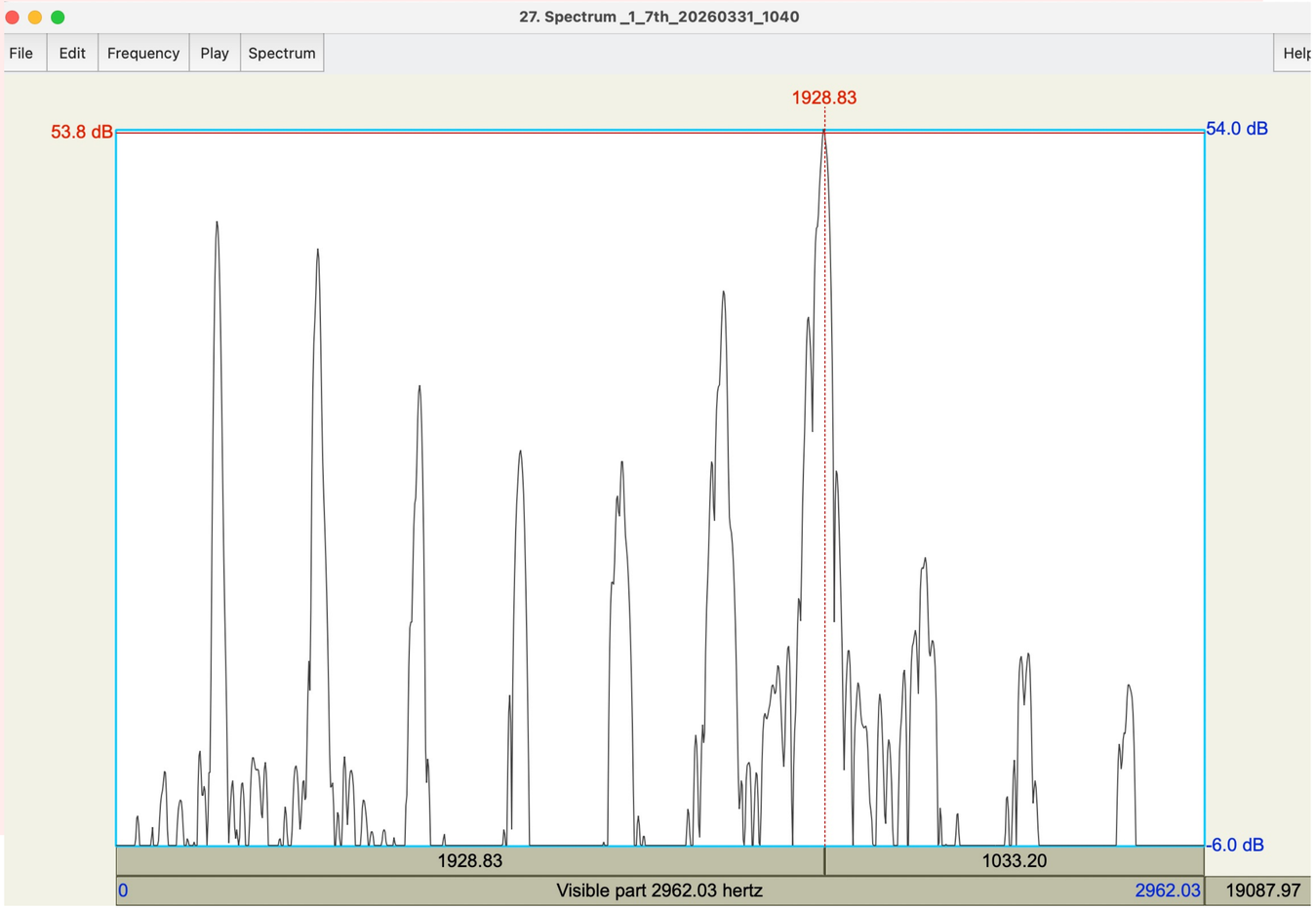
- Get median f0 across clip
- Get rms (intensity) of clip
- Get F1, F2, F3 (+ their bandwidths) from either the midpoint, or middle 50% of clip

Token	Vowel	f0	rms	F1	F2	F3	Duratic	BW1	BW2	BW3	Harmo	TargetF	LoNeigl	HiNeigl	TargetA	LoNeigl	HiNeigl	AvgNeig	TargetProm
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	1	269.9	269.9	269.9	48.06	48.06	48.06	48.06	0
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	2	539.8	269.9	809.7	44.58	48.06	31.92	39.99	4.592
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	3	809.7	539.8	1080	31.92	44.58	23.39	33.99	-2.07
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	4	1080	809.7	1349	23.39	31.92	23.51	27.72	-4.33
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	5	1349	1080	1619	23.51	23.39	27.95	25.67	-2.16
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	6	1619	1349	1889	27.95	23.51	24.74	24.13	3.823
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	7	1889	1619	2159	24.74	27.95	33.4	30.68	-5.93
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	8	2159	1889	2429	33.4	24.74	51.5	38.12	-4.73
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	9	2429	2159	2699	51.5	33.4	31.91	32.65	18.85
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	10	2699	2429	2969	31.91	51.5	19.62	35.56	-3.66
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	11	2969	2699	3239	19.62	31.91	19.24	25.57	-5.95
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	12	3239	2969	3509	19.24	19.62	28.9	24.26	-5.03
1	9th	269.9	0.133	512	2106	2432	1	195.9	629.9	9.182	13	3509	3239	3779	28.9	19.24	5.666	12.45	16.45
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	1	272.2	272.2	272.2	46.91	46.91	46.91	46.91	0
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	2	544.3	272.2	816.5	44.92	46.91	31.98	39.44	5.479
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	3	816.5	544.3	1089	31.98	44.92	24.06	34.49	-2.51
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	4	1089	816.5	1361	24.06	31.98	19	25.49	-1.43
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	5	1361	1089	1633	19	24.06	25	24.53	-5.53
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	6	1633	1361	1905	25	19	29.82	24.41	0.589
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	7	1905	1633	2177	29.82	25	50.3	37.65	-7.83
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	8	2177	1905	2449	50.3	29.82	24.78	27.3	23.01
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	9	2449	2177	2722	24.78	50.3	14.03	32.17	-7.39
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	10	2722	2449	2994	14.03	24.78	8.978	16.88	-2.85
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	11	2994	2722	3266	8.978	14.03	16.68	15.36	-6.38
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	12	3266	2994	3538	16.68	8.978	6.499	7.738	8.945
1	8th	272.2	0.131	529.9	1891	2177	1	169	630.8	11.8	13	3538	3266	3810	6.499	16.68	3.104	9.893	-3.39
1	7th	275.7	0.131	486.8	1665	1929	0.598	123.9	145.4	4.424	1	275.7	275.7	275.7	46.38	46.38	46.38	46.38	0
1	7th	275.7	0.131	486.8	1665	1929	0.598	123.9	145.4	4.424	2	551.4	275.7	827.1	44.07	46.38	32.58	39.48	4.584
1	7th	275.7	0.131	486.8	1665	1929	0.598	123.9	145.4	4.424	3	827.1	551.4	1103	32.58	44.07	27.15	35.61	-3.02
1	7th	275.7	0.131	486.8	1665	1929	0.598	123.9	145.4	4.424	4	1103	827.1	1378	27.15	32.58	26.42	29.5	-2.35
1	7th	275.7	0.131	486.8	1665	1929	0.598	123.9	145.4	4.424	5	1378	1103	1654	26.42	27.15	40.55	33.85	-7.43
1	7th	275.7	0.131	486.8	1665	1929	0.598	123.9	145.4	4.424	6	1654	1378	1930	40.55	26.42	54.09	40.26	0.293
1	7th	275.7	0.131	486.8	1665	1929	0.598	123.9	145.4	4.424	7	1930	1654	2206	54.09	40.55	18.21	29.38	24.71
1	7th	275.7	0.131	486.8	1665	1929	0.598	123.9	145.4	4.424	8	2206	1930	2481	18.21	54.09	10.19	32.14	-13.9
1	7th	275.7	0.131	486.8	1665	1929	0.598	123.9	145.4	4.424	9	2481	2206	2757	10.19	18.21	7.551	12.88	-2.69

Finding Resonance: Praat script

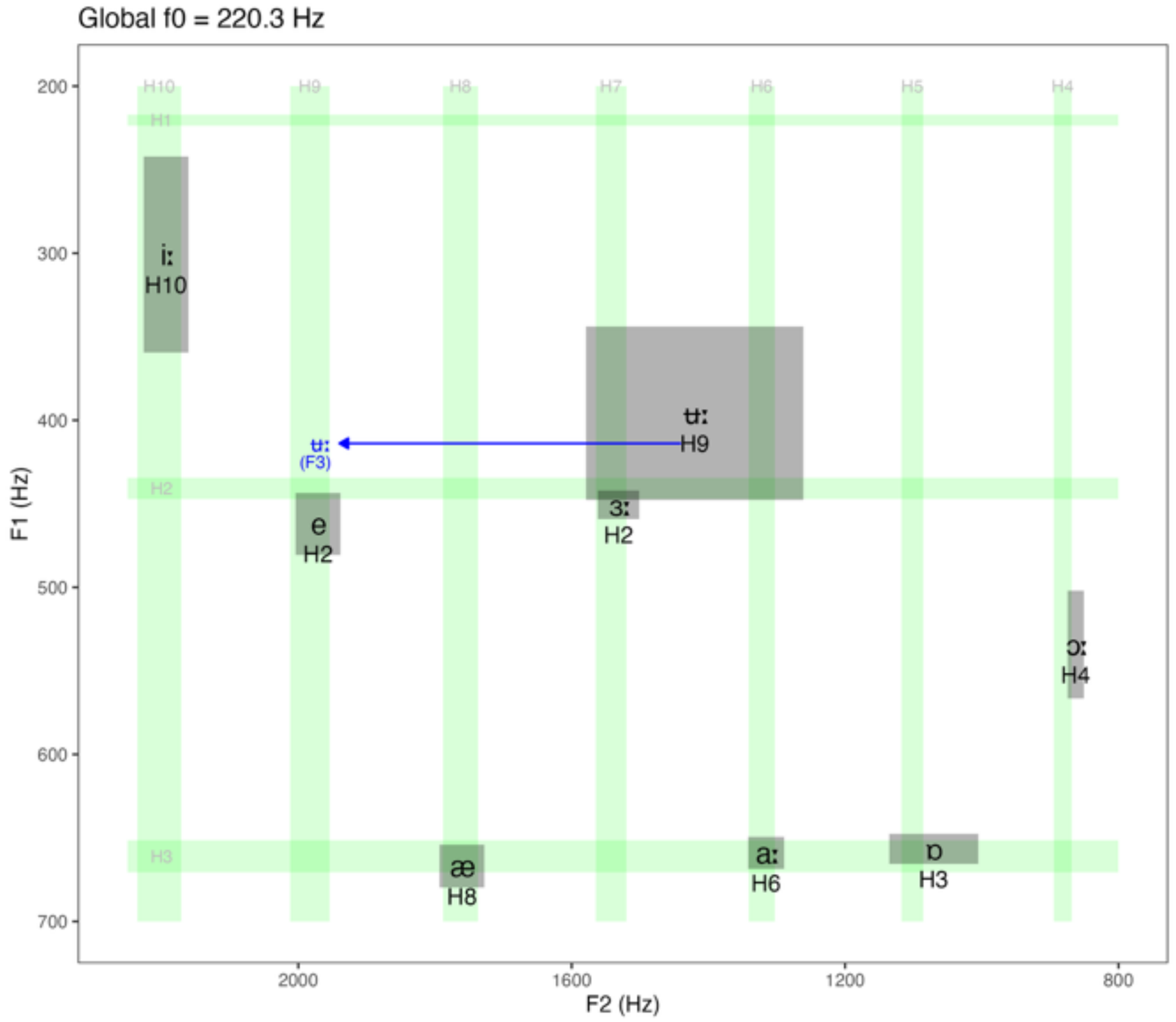
Analyse prominence of harmonics

- Create spectrum
- Measure amplitude at each harmonic of the fundamental
- Calculate average of the neighbours for each harmonic to determine relative prominence



Finding Resonance: R script

- Various modes (so far only single token)
- This example: stable f0, multiple vowels
- Harmonics overlaid on F1/F2 display
- F1/F2 of vowel tokens + bandwidths
- Most prominent harmonic labelled
- Arrow pointing to F3 when it is the nearest of F1/F2/F3 to a harmonic (in cents)



Summary

Singers manage linear and non-linear source-filter interactions

Many moving parts: every vowel will have ‘sweet spots’ and a ‘danger zones’ at every f0

Singers may unconsciously optimise vowel quality to: achieve resonance and vocal ease whilst avoiding f0 instability

Sociophonetic studies of singing need to understand singing technique!





Thanks to Jen Hay and Allan Bell
for guidance and collaboration over many years!